where, h is the Planck's constant (h = 6.63 x 10^{-34} J s) and number n takes natural numbers

(i.e., n = 1,2,3,....). Hence,

$$\mu_L = \frac{e}{2m} L = n \frac{eh}{4\pi m} A m^2$$

$$\mu_L = n \frac{(1.60 \times 10^{-19}) h}{4\pi m} A m^2$$

$$= n \frac{(1.60 \times 10^{-19})(6.63 \times 10^{-34})}{4 \times 3.14 \times (9.11 \times 10^{-31})}$$

$$\mu_L = n \times 9.27 \times 10^{-24} A m^2$$

The minimum magnetic moment can be obtained by substituting n = 1,

$$\mu_L = 9.27 \times 10^{-24} A m^2 = 9.27 \times 10^{-24} J T^{-1}$$
$$= (\mu_L)_{\min} = \mu_B$$

where, $\mu_B = \frac{eh}{4\pi m} = 9.27 \times 10^{-24} A m^2$ is called Bohr magneton. This is a convenient unit with which one can measure atomic magnetic moments.

Note: Bohr quantization rule will be discussed in unit 8 of second volume

3.9 AMPÈRE'S CIRCUITAL LAW

Ampère's circuital law is used to calculate magnetic field at a point whenever there is a symmetry in the problem. This is similar to Gauss's law in electrostatics. These are powerful methods whenever there is symmetry in the problem.

3.9.1 Definition and explanation of Ampère's circuital law

Ampère's law: The line integral of magnetic field over a closed loop is μ_0 times net current enclosed by the loop.

$$\oint_C \vec{B}.\vec{dl} = \mu_\circ I_{enclosed} \qquad (3.50)$$

where $I_{enclosed}$ is the net current linked by the closed loop C. Note that the line integral does not depend on the shape of the path or the position of the conductor with the magnetic field.

Note

Line integral means integral over a line or curve, symbol used is $\int$.

Closed line integral means integral over a closed curve (or line), symbol is $\oint$ *or* $\oint_c$

3.9.2 Magnetic field due to the current carrying wire of infinite length using Ampère's law

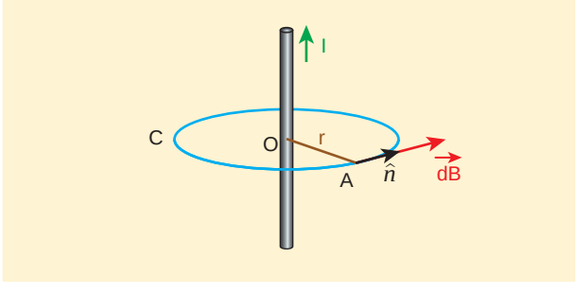


Figure 3.42 Ampèrian loop for current carrying straight wire

Consider a straight conductor of infinite length carrying current I and the direction of

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